

YOLO G.T. coordinates calculations

Given

$$\text{image size} = 70 \times 70$$

$$\text{grid size} = 7 \times 7$$

$$\text{cell size} = 70/7 = 10$$

Bounding Box center

$$(x_{\text{center}}, y_{\text{center}}) = (16, 44)$$

Bounding Box dimensions

$$w = 19, h = 34$$

Step 1Find grid cell containing ^{object} Center.

$$\text{column} = \left[\frac{16}{10} \right] = 1.6 = 1$$

$$\text{row} = \left[\frac{44}{10} \right] = 4.4 = 4$$

object lies in cell (1, 4)

use
floor
pass 1 but not
reached 2
pass 4 but not
reached 5.

step 2 find x & y relative position in cell.

$$x = \frac{x_{\text{center}} - x_{\text{cell-left}}}{\text{cell-width}}$$

$$y = \frac{y_{\text{center}} - y_{\text{cell-top}}}{\text{cell-height}}$$

$$x_{\text{cell-left}} = 1 \times 10 = 10$$

$$y_{\text{cell-top}} = 4 \times 10 = 40$$

step 3 calculate x & y

$$x = \frac{16 - 10}{10} = \frac{6}{10} = 0.60$$

$$y = \frac{44 - 40}{10} = \frac{4}{10} = 0.40$$

step 4 width & height (relative to image)

$$w = \frac{\text{box width}}{\text{image width}}$$

$$h = \frac{\text{box height}}{\text{image height}}$$

$$w = \frac{19}{70} = 0.27$$

$$h = \frac{34}{70} = 0.49$$

final ans

$$(x, y, w, h) = (0.60, 0.40, 0.27, 0.49)$$

practice 8

image size 84×84

divided into 7×7 grid

center $(53, 18)$

width 21

height 14.

find : (1) Grid cell (row, col) containing
obj center

(2) normalized x, y, w, h .

Ans:

row = 1, col = 4.

$$(0.42, 0.50, 0.25, 0.17)$$